

Shengfan Cao

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Summary

UC Berkeley Ph.D. candidate developing safe, machine-learning-based prediction, planning, and control systems for autonomous vehicles. First-author IEEE IROS oral presenter with 4+ years across machine learning, planning, and controls, spanning PyTorch research, CARLA evaluation, embedded-robot deployment, and C++ production planner integration.

Technical Skills

- **Machine learning:** PyTorch, deep learning, reinforcement learning, imitation learning, actor-critic methods, Diffusion Transformers (DiT), domain adaptation, CNNs, safe reinforcement learning; NumPy, Pandas.
- **Planning and safety:** Trajectory optimization, MPC, DDP, reachability analysis, constrained optimization, optimal control, autonomous driving.
- **Programming and systems:** Python, C++17, MATLAB, SQL; CARLA, ROS 1/2, CasADi, Acados, Bazel, Git, Pytest, Linux, Docker, NVIDIA Jetson.

Education

University of California, Berkeley

Aug 2022 — May 2027

Ph.D. Candidate, Mechanical Engineering; PI: Francesco Borrelli

- **Coursework:** Deep Learning, Reinforcement Learning, Computer Vision, Model Predictive Control, Convex Optimization, Nonlinear Systems, Natural Language Processing.

Tsinghua University

Aug 2017 — Jun 2022

B.S. in Mechanical Engineering; B.A. in Japanese Language and Literature

Autonomous Driving and Production Experience

Didi Research America

May 2026 — Aug 2026

Motion Planning Engineer, Intern

- Collaborated with the ML team, implemented and integrated a DDP smoother into the training pipeline of an end-to-end motion planning model, using large-scale human driving datasets (1.3B frames), achieving superior comfort and efficiency compared to the baseline in a data-based simulation.
- Redesigned and improved the optimization-based trajectory preprocessor as a Bayesian MAP inference, solved with augmented-Lagrangian DDP, reducing jerk and lateral-jerk discomfort rates to 0–4% from 8–39%, while keeping ADE/FDE competitive.
- Developed an MPC-based motion planner for autonomous U-turns / K-turns in narrow intersections, automatically selecting feasible maneuvers online.
- Proposed and evaluated a novel dual-learning framework for constrained optimization, improving solve speed by 6x over an IPOPT baseline.
- **Tech:** C++17, Python, CasADi, Acados/HPIPM/BLASFEO, IPOPT, Eigen, pybind11, ROS 1, Bazel.

Selected Publications

- Cao, S., Joa, E., & Borrelli, F. (2025). *A Simple Approach to Constraint-Aware Imitation Learning with Application to Autonomous Racing* [🔗](#). **Oral presentation** at IEEE IROS 2025.
- Cao, S., Borrelli, F., & Joa, E. (2025). *Constrained Policy Optimization via Sampling-Based Weight-Space Projection* [🔗](#). **Oral presentation** at IFAC World Congress 2026.
- Cao, S., & Liu, Y. (2025). *State-Conditional Adversarial Learning: An Off-Policy Visual Domain Transfer Method for End-to-End Imitation Learning* [🔗](#). **Interactive presentation** at IFAC World Congress 2026.
- Prignoli, F., Cao, S., Falcone, P., & Borrelli, F. (2025). *Real-Time Regulation-Aware Game-Theoretic Motion Planning for Head-to-Head Autonomous Racing*. Under review at IEEE TCST.

Machine Learning Research

Safe Imitation Learning at Handling Limits, Principal Researcher

Nov 2024 — Feb 2025

- Designed a constraint-aware imitation-learning framework with an actor-critic architecture for explicit safety modeling.
- Integrated reachability-based safety filters into the training pipeline to support policy generalization beyond the training distribution.
- Improved training efficiency by 150% and reduced collision rates by 60% in CARLA relative to unconstrained baselines.
- Evaluated success rates and failure modes under safety-critical conditions near the vehicle's dynamic handling limits.

Sampling-Based Constrained Policy Optimization, Principal Researcher

Sep 2025 — Dec 2025

- Introduced a scalable, LoRA-inspired, training-time constrained-optimization framework for large neural policies without test-time shielding or policy rollback.
- Formulated neural-policy updates as a convex projection in parameter space, with theoretical guarantees that projected updates preserve objective improvement.
- Derived a sufficient stability condition connecting weight updates to closed-loop constraint satisfaction and embedded it into the optimization pipeline.
- Demonstrated complete rejection of harmful supervision and safe performance gains under adversarial experts.

Safe Diffusion Policies for Robot Learning, Principal Researcher

Mar 2026 — Present

- Investigating inference-time safety mechanisms for diffusion policies in long-horizon robot decision making.
- Built a PyTorch training and evaluation framework for Diffusion Transformer (DiT) policies across navigation and manipulation domains.
- Designed experiments to analyze safety-performance tradeoffs and closed-loop consistency under different planning formulations.
- Demonstrated encouraging preliminary results in path planning while identifying key challenges for receding-horizon control and manipulation.

Trusted Autonomy for Uncrewed Ground Vehicles, Project Lead

Sep 2024 — Present

- Developed an interpretable autonomy stack that converts multimodal sensor data into human-readable terrain and uncertainty maps.
- Maintained a continuously updated latent world representation combining geometry, semantics, and uncertainty for long-horizon planning.
- Deployed the PyTorch/ROS 2 autonomy stack on Jetson Orin under onboard compute constraints and collaborated with NIWC Pacific researchers.
- Built a Python/SQLite access layer for querying and analyzing logged ROS 2 autonomy data.

Software, Teaching, and Leadership

Vehicle-Dynamics Simulation Infrastructure

Jan 2025 — May 2025

Graduate Student Instructor, UC Berkeley

- Built a dynamic-bicycle simulator with Gymnasium APIs, a CARLA connector, and automated installation, used as engineering-course infrastructure.

Supervisor for Undergraduate Research Program, UC Berkeley

Jan 2025 — Present

- Led four students on vision-based off-road vehicle-dynamics modeling and path planning, including documentation and reviews with faculty and industry stakeholders.

Invited Reviewer, IEEE and IFAC Venues

Apr 2025 — Present

- Reviewed seven submissions across Automatica, CDC, IFAC World Congress, IEEE CASE, and IEEE Robotics and Automation Letters.